

Task 3 Report

Task 3: Sales Prediction Model

Introduction

The objective of this task was to develop machine learning models capable of predicting sales values using historical e-commerce transaction data.

Dataset Description

The cleaned dataset from Task 1 was used as the input dataset. The target variable selected for prediction was `total_amount`.

Feature Engineering

Several new features were created to improve predictive performance:

Date-Based Features

The order date column was converted into datetime format and the following features were extracted:

- Year
- Month
- Day
- Day of Week

Lag Feature

A previous sales feature was created using historical sales values to capture sales continuity.

Rolling Average Feature

A rolling average sales feature was generated using a seven-day moving window to capture sales trends.

Data Preprocessing

Feature Selection

Non-essential columns such as order ID, customer ID, product ID, and order date were removed from the feature set.

Encoding Categorical Variables

Categorical variables were transformed using one-hot encoding to make them suitable for machine learning algorithms.

Train-Test Split

The dataset was divided into:

- Training Data: 80%
- Testing Data: 20%

Model Development

Linear Regression

A Linear Regression model was implemented as a baseline model for sales prediction.

Random Forest Regressor

A Random Forest Regressor model was trained to capture complex patterns and non-linear relationships in the data.

Model Evaluation

The models were evaluated using:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- R-Squared Score (R^2)

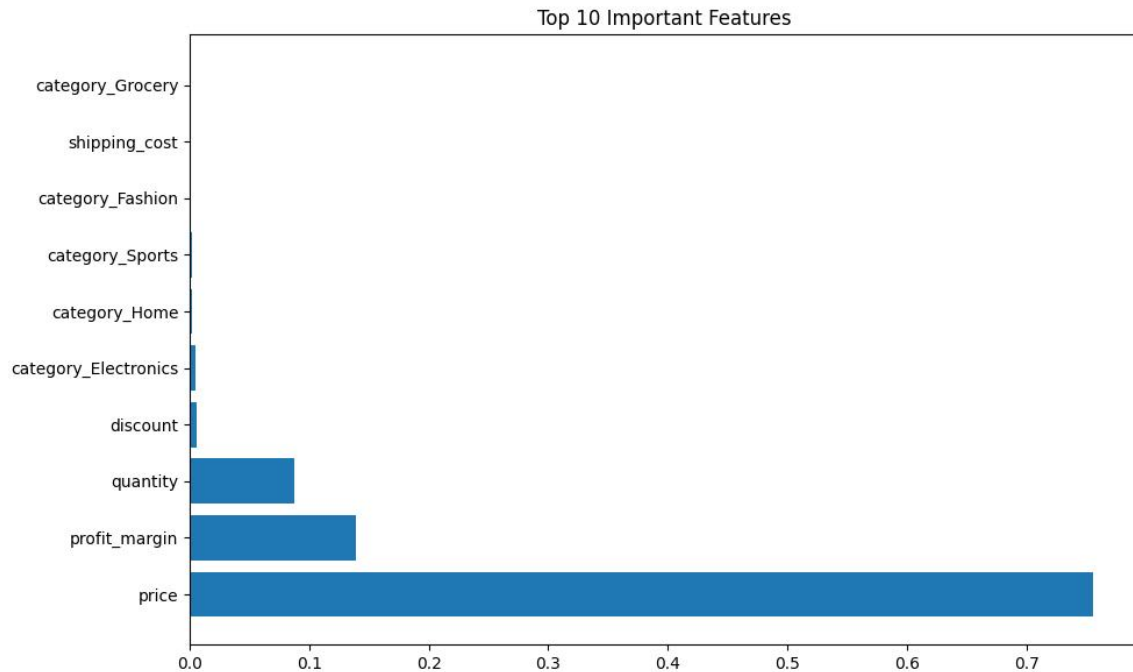
A comparison table was generated to compare the performance of both models.

Hyperparameter Tuning

GridSearchCV was used to optimize Random Forest parameters and identify the best-performing model configuration.

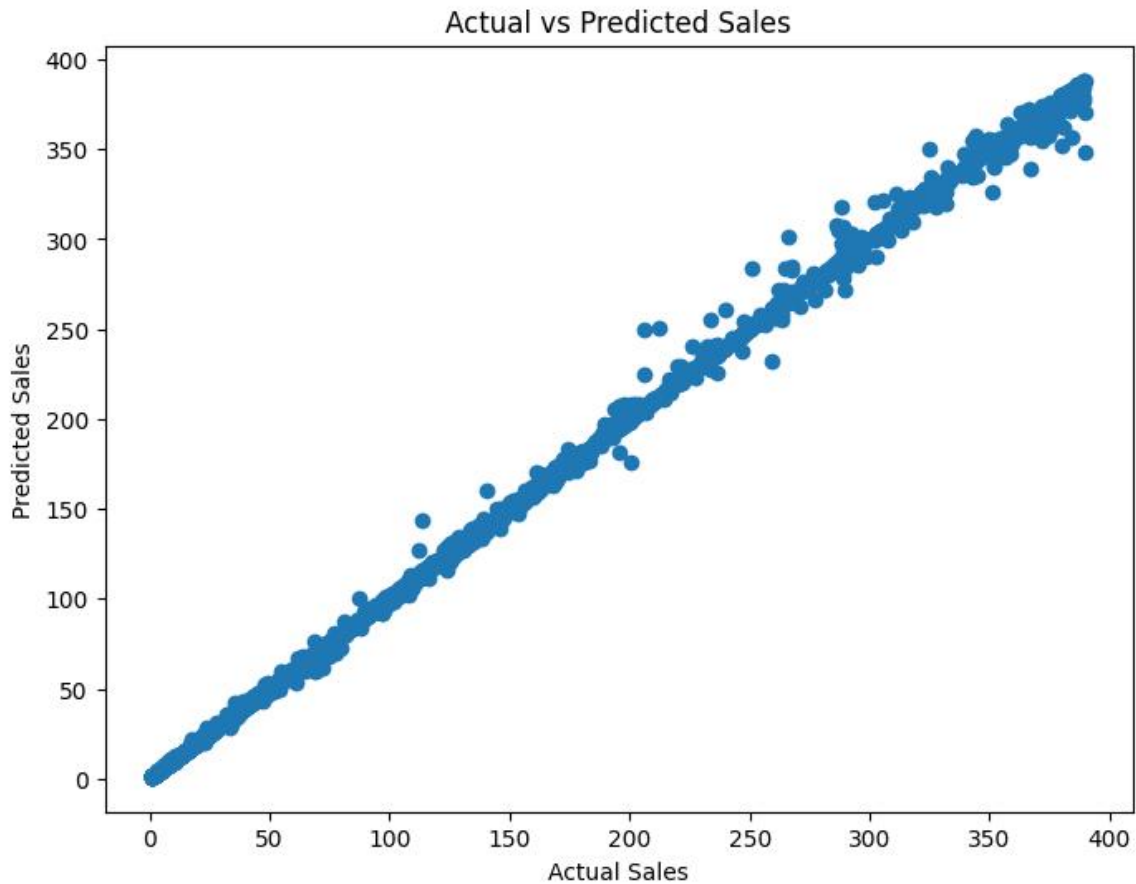
Feature Importance Analysis

Feature importance scores were calculated and visualized to identify the variables that contributed most to sales prediction.



Actual vs Predicted Analysis

A scatter plot comparing actual and predicted sales values was created to evaluate model performance visually.



Business Interpretation

- Historical sales behavior significantly influences future sales.
- Product price and quantity are important factors affecting sales performance.
- Predictive modeling can assist in demand forecasting and inventory planning.

Conclusion

The machine learning pipeline was successfully implemented. Linear Regression and Random Forest models were trained and evaluated, with Random Forest providing superior predictive performance. The final model can be used to support business forecasting and decision-making processes.